Information-Triggered Delivery for Timely Victim Reporting in Infrastructure-Free UAV Search and Rescue

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***Abstract*—Small UAV swarms with onboard thermal detectors can search disaster areas without infrastructure, but a victim found by a UAV helps only once the report reaches the responders, over radio links that are obstructed, fading and broken by UAV failures. Keeping the swarm connected at all times makes every report immediate but confines the search; searching freely leaves delivery to chance. We present SENTINEL, a decentralised framework that lets UAVs search freely and triggers delivery from the information each UAV has confirmed, fuses thermal detections from several viewpoints so that occluded angles cannot veto a clear one, and, while a report is being delivered, constrains the UAVs on the delivery path with an outage model and a local minimum spanning tree so that preserved links remain reliable for every feasible step. We bound the probability that a confirmed detection is lost before it is secured and show that nearest-neighbour distance constraints do not preserve connectivity. In about 11 000 simulated missions against seven baselines tuned with the same budget (connectivity control, potential fields, communication-aware PSO, MAPPO and sweep search), with 30 seeds per setting and paired tests, SENTINEL reports 23–30 % more victims within 60 s than the best baseline in urban rubble under six failure models, is significantly better than all baselines for 3–50 UAVs on a 400 m area, and is never significantly worse; PX4 software-in-the-loop flights in Gazebo agree with the fast simulator. Ablations attribute the gains to the delivery trigger and the fusion rule, not to continuous connectivity. The onboard YOLO26n detector matches YOLOv8n in accuracy on HIT-UAV with 31 % less compute.**

***Index Terms*—UAV swarms, intermittent connectivity, search and rescue, edge AI, particle swarm optimization, delay-tolerant networking.**

I. Introduction

After earthquakes, floods and large fires, cellular and backhaul infrastructure is often unavailable precisely where responders need information. Small multirotor UAVs carrying thermal cameras can search such areas quickly, and onboard detectors now run in a few milliseconds on embedded GPUs, so victims can be detected without streaming video to a server [1]. What remains hard is getting each detection to the responders *in time*. Without infrastructure, a report travels only over the UAVs' own short-range radio links, which are obstructed by buildings, fade, and break when UAVs fail.

Two families of methods address the communication side. Connectivity-maintenance controllers keep the swarm's communication graph connected at all times, either through the algebraic connectivity of the graph [2], [3] or through edge potentials that never let a link break [4]. Every detection then reaches the responder immediately, but the whole swarm is tied to the responder's neighbourhood, which limits how far it can search. Intermittent-connectivity and data-ferrying approaches let robots disconnect and meet according to a schedule [5], or plan ferry routes to sources whose positions are known in advance [6], [7]. In search and rescue neither assumption fits well: the information sources, the victims, are *discovered by the search itself*, at unknown places and times.

In an earlier version of this system we coupled a thermal detector with particle swarm optimisation (PSO) and a distance penalty that kept UAVs within 50 m of their nearest neighbour. That design has three limitations, which also shaped this work. A fixed distance does not describe a radio link behind rubble; keeping the nearest neighbour close does not keep the network connected (Lemma 1); and the evaluation used a 25 m area that is far smaller than the radio range, in which, as we show in Section VI-J, every method stays connected and the methods cannot be told apart.

This paper studies the *discovery–delivery trade-off* directly. We measure the share of victims whose report reaches the responder within a deadline, and we treat connectivity as a means to that end rather than as an objective. Our contributions are:

- **Information-triggered delivery.** Each UAV searches without connectivity constraints and switches to a delivery mode when a victim it confirmed has not reached the responder after a hold time T_h (Section IV). Rendezvous are triggered by information that the search itself discovers, rather than by a schedule or by known source locations. T_h exposes the trade-off explicitly, and we bound the probability that a confirmed detection is lost to UAV failures before it is secured (Proposition 3).
- **Corroborative multi-view sensing.** Weak detections request a lateral second view from a neighbour, and views from different directions are fused so that occluded viewpoints cannot veto a clear one (Section IV-G). In dense swarms this fusion rule is the largest single contributor to performance.
- **A link-preservation safety layer with guarantees.** While a delivery is in progress, the UAVs on the delivery path constrain their steps using a realistic outage model (path loss, obstructions, shadowing, fading, interference) and a local minimum spanning tree. Preserved links remain reliable for every feasible step (Proposition 1), and the delivering component stays connected (Corollary 1). We show why nearest-neighbour constraints do not suffice (Lemma 1).
- **A reproducible evaluation.** We compare against seven

baselines covering communication-aware optimisation, algebraic-connectivity control, potential-based topology control, multi-agent reinforcement learning and classical sweep search, all tuned with the same budget. The evaluation spans 3–50 UAVs, four channel profiles, six failure models, moving obstacles and detector uncertainty, uses 30 seeds per configuration with paired significance tests, and is cross-checked against PX4 software-in-the-loop flight in Gazebo. Code, scenarios and results are released.

Across the scenarios SENTINEL reports 23–30 % more victims within 60 s than the best baseline in urban rubble under all failure models, and it is significantly better than all baselines on the 400 m area at every swarm size from 3 to 50; in the harshest channel it is ahead of or tied with the best baseline. The ablation shows where this comes from: the delivery trigger and the fusion rule account for the gains, whereas the link-preservation layer has no significant effect on performance and fulfils its safety role, with 3.4×10^{-5} link violations per preserved link-step. We also report what does not work as well: keeping the swarm connected at all times lowers on-time reporting to 0.03–0.13; in small, dense swarms a MAPPO policy trained to convergence comes close to SENTINEL (differences not significant for $N = 10$–20 on the 200 m area); and the thermal detector, trained on real imagery, transfers poorly to rendered Gazebo scenes.

Section II positions this work against prior approaches, Section III defines the system model and problem, Section IV presents SENTINEL, Section V describes the implementation, Section VI reports the evaluation and Section VII discusses limitations.

II. Related Work

a) Continuous connectivity maintenance: A large body of work keeps a multi-robot communication graph connected at all times. Zavlanos and Pappas control the robots so that the graph Laplacian keeps a positive second eigenvalue [3]; Sabattini et al. estimate the algebraic connectivity λ_2 in a decentralised way and follow its gradient without requiring local links to persist [2]; Ji and Egerstedt preserve edges through potentials that grow unbounded as a link approaches its range, with hysteresis on edge creation [4]. Local topology-control rules such as the local minimum spanning tree (LMST) [8] reduce the set of links that must be kept. These methods prove that connectivity is maintained, but under a disk communication model and at the price of tying every robot to the network at every instant. Our baselines B3 and B4 implement the second and third approach (B3 with the exact λ_2), and our measurements show that in large areas this tie lowers the share of victims reported in time. SENTINEL applies such constraints only for the duration of a delivery (Proposition 1) and replaces the disk model by a predicted outage model that accounts for obstructions and fading.

b) Intermittent connectivity, data ferrying and information age: Kantaros and Zavlanos let robots disconnect and enforce that they meet at the vertices of a mobility graph infinitely often, with meeting schedules specified in temporal logic [5]. UAV data-ferrying and delay-tolerant designs carry data physically between sources and a sink [6], [9], and age-of-information (AoI) formulations optimise how fresh the collected data are [7], [10], [11]. These approaches assume that the sources (sensors) are known in advance or that meetings follow a predefined schedule. In search and rescue the sources are the victims, discovered online by the search itself; SENTINEL's rendezvous are triggered by that discovered information, and our deadline metric is AoI-like (time from confirmation to the responder).

c) Communication-aware swarm search and learning: PSO and its variants are widely used for multi-UAV coverage and search [12], [13]; distributed planners such as DAPSO run a local PSO on each UAV and exchange information over local links, but assume that communication is available when needed [14]. Communication-aware variants add link-quality terms to the fitness; our baseline B2 represents this family with the same outage model as SENTINEL applied as a soft penalty to all neighbours. Distributed Bayesian search reduces traffic by flooding only positive detections but assumes a connected, lossless network [15], and decentralised mountain-SAR search uses distance-limited links without a responder sink [16]. Multi-agent reinforcement learning, with MAPPO as a strong general baseline [17], has been applied to UAV coverage and communication-aware coordination [18]; we include MAPPO with the same observations and objective as baseline B5.

d) Onboard thermal perception: Thermal datasets such as HIT-UAV [1] enable lightweight detectors that run on the UAV, and recent YOLO generations target edge deployment by removing non-maximum suppression and reducing compute [19]. We use such a detector as an enabling component and evaluate it on real thermal imagery; we do not claim it as a contribution.

e) Positioning: Table I summarises the comparison. To our knowledge, no prior work combines online-discovered information sources, an explicit reporting deadline, a channel model with obstructions and fading, failure-aware evaluation, and provable link preservation during delivery (Proposition 1).

III. System Model and Problem Formulation

A. Network and Mobility

A swarm $\mathcal{U} = \{1, \dots, N\}$ of multirotor UAVs flies at a common altitude h over a disaster area $\mathcal{A} \subset \mathbb{R}^2$ containing static obstacles. Obstacles taller than h form a no-fly set $\mathcal{O} \subset \mathcal{A}$ and obstruct radio links; low rubble only occludes ground targets. A responder radio (node 0) is fixed at the launch point $x_0 \in \mathcal{A}$; it is the only sink for victim reports and is assumed not to fail. UAV i has position $x_i(t) \in \mathcal{A} \setminus \mathcal{O}$ and speed $\|\dot{x}_i\| \le v$. Time is slotted with control period Δt; every UAV broadcasts one beacon per slot containing its position, its target ledger and acknowledgements (Section IV).

B. Channel Model

The received SINR of the broadcast from i at j in slot t is

$$\gamma_{ij}(t) = \frac{P_t G^2 \, 10^{-(\mathrm{PL}(d_{ij}) + n_{ij} L_o + X_{ij})/10} \, |h_{ij}|^2}{\sigma^2 + \sum_{k \in \mathcal{I}_j(t)} \bar{P}_{kj}}, \quad (1)$$

TABLE I
COMPARISON WITH THE CLOSEST APPROACHES (✓: ADDRESSED; ○: PARTIALLY; –: NOT ADDRESSED).

Approach	Sources discovered online	Delivery deadline / timeliness metric	Obstruction/fading channel model	UAV failures	Connectivity proof (Prop. 1)	Free search (no permanent tie)
λ_2 control [2], [3]	–	–	–	○	✓	–
Edge potentials [4]	–	–	–	–	✓	–
Intermittent connectivity [5]	–	○	–	–	✓	✓
Data ferrying / DTN [6], [9]	–	✓	○	–	–	✓
AoI UAV data collection [7], [10], [11]	–	✓	○	–	○	✓
Distributed PSO search [13], [14]	✓	–	–	–	–	✓
Event-driven Bayesian search [15]	✓	–	–	–	–	✓
MARL coordination [17], [18]	○	○	○	○	–	✓
Earlier SENTINEL design (B1)	✓	–	–	○	–	○
SENTINEL (this work)	✓	✓	✓	✓	✓†	✓

† during delivery, for feasible steps (Proposition 1, Corollary 1).

TABLE II
CHANNEL PROFILES ($f = 2.4\,\text{GHz}$, $P_t = 20\,\text{dBm}$, $B = 20\,\text{MHz}$, $\gamma_{\text{th}} = 5\,\text{dB}$) AND RESULTING RELIABLE RANGES AT $\varepsilon = 0.05$.

Profile	η	σ_X	K (LoS)	L_o	$R(0)$	$R(1)$
open	2.0	2	10 dB	6 dB	2434 m	684 m
suburban	2.5	4	6 dB	10 dB	339 m	95 m
urban-rubble	3.0	6	3 dB	6 dB	91 m	48 m
severe	3.5	8	Rayleigh	8 dB	35 m	21 m

where $\text{PL}(d) = \text{PL}(d_0) + 10\eta \log_{10}(d/d_0)$ is log-distance path loss, n_{ij} the number of obstacles crossing the segment $[x_i, x_j]$ with per-obstruction loss L_o, $X_{ij} \sim \mathcal{N}(0, \sigma_X^2)$ log-normal shadowing evolving with the Gudmundson correlation $\rho = e^{-\Delta/d_c}$ for endpoint displacement Δ, $|h_{ij}|^2$ unit-mean Rician fading with factor K ($K = 0$, Rayleigh, once obstructed), and $\mathcal{I}_j(t)$ the set of transmitters colliding in the slot, each present with probability p_a. A packet is decoded iff $\gamma_{ij} \geq \gamma_{\text{th}}$. The link outage probability, averaged over fading and shadowing, is

$$P_{\text{out}}(d, n) = \mathbb{E}_X \left[F_{|h|^2} \left(\frac{\gamma_{\text{th}}}{\bar{\gamma}(d, n)\, 10^{X/10}} \right) \right], \qquad (2)$$

with $F_{|h|^2}$ the non-central χ_2^2 CDF (computed in closed form; the outer expectation by Gauss–Hermite quadrature). The *reliable range* for target outage ε is $R_\varepsilon(n) = \max\{d : P_{\text{out}}(d, n) \leq \varepsilon\}$, and the *reliable graph* $\mathcal{G}_\varepsilon(t)$ joins i, j iff $P_{\text{out}}(d_{ij}, n_{ij}) \leq \varepsilon$. Table II lists the four environment profiles; their parameters follow the ranges reported for low-altitude aerial and street-level links [20], [21], with shadowing correlation following [22]; they are not calibrated against measurements in rubble (Section VII). For the urban-rubble profile $R_{0.05}(0) = 91\,\text{m}$ and $R_{0.05}(1) = 48\,\text{m}$, which places the fixed 50 m range of the earlier SENTINEL design in context: it is the reliable range of a link behind one obstruction, not a property of the radio alone.

C. Sensing Model

Each UAV carries a nadir thermal camera with ground footprint radius r_f and an onboard detector (Section V). A look at a cell updates the residual probability that an undetected victim is present by Bayes' rule, $p \leftarrow p(1 - P_d)/(1 - pP_d)$ [23]. The detector returns a confidence $c \in [0, 1]$ whose distribution depends on slant range and on view-dependent occlusion (calibrated from the detector, Section VI); false positives from warm debris arrive as a Poisson process per look.

D. Information Model

Every UAV keeps a ledger of world-frame target entries. Observations within a gate are associated to one entry; the best confidence per viewing sector is kept and the sectors are fused into one confidence c_e (Section IV-G). An entry is *confirmed* when $c_e \geq c_{\text{hi}}$ and *tentative* when $c_{\text{lo}} \leq c_e < c_{\text{hi}}$. Ledgers are replicated by gossip with acknowledgements, so each entry carries its holder set $H_e(t)$. An entry is *secured* once $0 \in H_e$ (the responder holds it) and *lost* if it was confirmed but no alive node holds it.

E. Failure Model

UAVs fail by crashing (the node and its ledger copy disappear), by radio loss (it flies but cannot communicate) or by gradual brownout (transmit power decays). We study single, simultaneous, cascading and Poisson failures with per-UAV rate λ_f.

F. Problem

Let τ_k be the time victim k is first confirmed by any UAV and $T_k \geq \tau_k$ the time its report reaches the responder ($T_k = \infty$ if never). With mission horizon T and value discount θ, the team maximises the time-discounted report value

$$J = \frac{1}{|\mathcal{V}|} \sum_{k \in \mathcal{V}} e^{-T_k/\theta}, \qquad (3)$$

subject to the dynamics, $x_i \notin \mathcal{O}$, and decentralised information (each UAV acts on its own sensing and on beacons it actually received). J couples three quantities that prior work optimises separately: discovery (a victim must be found), delivery (its report must reach the responder), and timeliness (sooner is better, in the spirit of age of information [10]). Maintaining $\mathcal{G}_\varepsilon$ connected at all times makes delivery immediate but restricts discovery; pure exploration maximises discovery but leaves delivery to chance contacts, so confirmed information can wait arbitrarily long or be lost with its carrier. We refer to this tension as the *discovery–delivery trade-off*.

IV. SENTINEL

Each UAV runs the same decentralised agent (Algorithm 1), built from three parts. (i) *Information-triggered delivery*: the agent alternates between two modes driven by its own information state; in EXPLORE it searches without any connectivity constraint, in DELIVER it returns toward the responder until the victims it has confirmed are secured. (ii) *Corroborative multi-view sensing*: weak detections request a second viewpoint from a neighbour, and views are fused so that occluded angles cannot veto a clear one. (iii) *A link-preservation safety layer*: while delivering, a UAV and the relays between it and the responder constrain their motion so that the links they rely on stay reliable, with a guarantee for feasible steps (Proposition 1). Section VI-H shows that (i) and (ii) produce the performance gains, while (iii) ensures that deliveries are not interrupted by the swarm's own motion.

A. Mode switching

Let $\mathcal{E}_i(t)$ be the confirmed entries held by UAV i that are not secured and for which i is responsible (it created them, or the creator has been silent for $3T_h$, i.e. the copy is orphaned). UAV i switches to DELIVER when the oldest such entry reaches age T_h,

$$\max_{e\in\mathcal{E}_i(t)} (t-\tau_e) \;\geq\; T_h, \tag{4}$$

and back to EXPLORE when $\mathcal{E}_i(t)=\emptyset$. The hold time T_h is the single parameter that trades discovery against delivery latency: $T_h = 0$ delivers every confirmation immediately, $T_h \to \infty$ never interrupts the search. Whether a UAV is connected to the responder is known locally from a heartbeat: the responder stamps its beacons with the current time and every UAV relays the freshest stamp it received; UAV i is *connected* if its freshest stamp is younger than T_{hb}.

B. Waypoint optimisation (EM-PSO)

Every planning period the agent chooses its next waypoint inside the reachable disk $\mathcal{D}_i = \{x : \|x - x_i\| \leq vT_p\} \setminus \mathcal{O}$ by maximising

$$F_i(x) = w_c\,\hat{G}_i(x) + w_d \sum_{e\in\mathcal{T}_i} c_e\, e^{-\frac{\|x-x_e\|^2}{2s^2}} + w_g\,\frac{\|x_i-g_i\| - \|x-g_i\|}{vT_p} - S_i(x), \tag{5}$$

where $\hat{G}_i(x)$ is the normalised expected number of detections of a look at x computed on the UAV's own search map (cells inside footprints claimed by neighbours removed), $\mathcal{T}_i$ the tentative entries, and g_i a frontier goal (the map cell maximising expected detections discounted by distance). The *signal spring* acts only while the UAV is tethered (in DELIVER and connected, or holding a relay tether, Section IV-D):

$$S_i(x) = \lambda_i \sum_{j\in\mathcal{S}_i} \left(\frac{[P_{\mathrm{out}}(\|x-x_j\|, n(x,x_j)) - \varepsilon_i]_+}{\varepsilon_i} \right)^2, \tag{6}$$

with $\lambda_i = \lambda_0(1+\beta U_i)$ and $\varepsilon_i = \varepsilon/(1+\beta U_i)$, where U_i is the number of entries only i holds. Unlike a distance penalty, (6) penalises the predicted *outage* of the link, so it accounts for obstructions and fading, and it tightens with the amount of unique information the UAV carries. The maximisation uses a constriction-coefficient PSO ($\chi = 0.7298$, $c_1 = c_2 = 2.05$) [12], whose particles are seeded with the previous waypoint, the step toward g_i and the waypoints broadcast by neighbours.

C. Link preservation (safety layer)

The spring (6) steers the search toward reliable links but cannot ensure that they survive. While tethered, UAV i therefore projects its step onto a feasible set built from its preserved neighbours $\mathcal{S}_i$.

Definition 1 (Local MST neighbours)**.** Let $\hat{\mathcal{G}}_i$ be the subgraph of the reliable graph induced by i and the neighbours it heard within the last T_{hb}, with edge weights $(d_{jk}, \max(j,k), \min(j,k))$ ordered lexicographically. $\mathcal{S}_i$ is the set of neighbours adjacent to i in the minimum spanning tree of $\hat{\mathcal{G}}_i$ [8].

Lemma 1 (Nearest-neighbour constraints do not imply connectivity)**.** *There are configurations in which every UAV is within R of its nearest neighbour while the communication graph is disconnected.*

Proof. Four UAVs at $(0,0),(10,0),(200,0),(210,0)$ with $R = 50\,\mathrm{m}$: each has a neighbour at 10 m, yet no edge joins $\{1,2\}$ to $\{3,4\}$. □

Lemma 1 is why the nearest-neighbour spring of the earlier SENTINEL design cannot certify network maintenance.

Lemma 2 (LMST spans the reliable graph [8])**.** *Let $\mathcal{G}^-$ contain edge (i,j) iff $j\in\mathcal{S}_i$ and $i\in\mathcal{S}_j$, all computed from the same positions. Then $\mathcal{G}^-$ contains the minimum spanning forest of the reliable graph, hence has the same connected components.*

Proof. Let (i,j) be an edge of the global minimum spanning forest $\mathcal{M}$. If (i,j) were not in the local MST of $\hat{\mathcal{G}}_i$, the cycle property would give a cycle in $\hat{\mathcal{G}}_i$ on which (i,j) is the unique heaviest edge; that cycle exists in the reliable graph, contradicting $(i,j)\in\mathcal{M}$. The same holds for $\hat{\mathcal{G}}_j$, so $(i,j)\in\mathcal{G}^-$. Since $\mathcal{M}\subseteq\mathcal{G}^-$ and $\mathcal{G}^-$ only uses reliable links, the components coincide. □

a) Midpoint-ball projection: For each $j\in\mathcal{S}_i$ let $\hat{x}_j$ be j's last reported position, of age a_{ij}, $m_{ij} = (x_i+\hat{x}_j)/2$, and $\delta_{ij} = v(\Delta t + a_{ij})$ the furthest j can move before the next check. The proposed step u_i is scaled to the largest $\alpha\in[0,1]$ such that $x_i' = x_i + \alpha u_i$ satisfies, for all $j\in\mathcal{S}_i$,

$$\|x_i' - m_{ij}\| \leq \tfrac{1}{2} r_{ij}(x_i') - \tfrac{1}{2} v\, a_{ij}, \quad r_{ij}(x_i') = R_\varepsilon\big(n^{\delta_{ij}}(x_i', \hat{x}_j)\big), \tag{7}$$

where $n^\delta(a,b)$ counts obstacles inflated by δ that cross $[a,b]$. If no $\alpha>0$ satisfies (7) the UAV hovers, or, if it is already outside a constraint, takes the largest step toward the violated centre that does not enlarge any other violation. Each UAV evaluates (7) with local information only.

Proposition 1 (Link preservation)**.** *Suppose $j\in\mathcal{S}_i$ and $i\in\mathcal{S}_j$, both UAVs are tethered, speeds are bounded by v, positions*

are exchanged with staleness at most a, and the steps of both UAVs satisfy (7)*. Then the link (i, j) is ε-reliable at $t + \Delta t$.*

Proof. Let $m = (x_i + x_j)/2$ be the true midpoint. Since $\|\hat{x}_j - x_j\| \le v a_{ij}$, we have $\|m_{ij} - m\| \le v a_{ij}/2$ and, symmetrically, $\|m_{ji} - m\| \le v a_{ji}/2$. By (7) and the triangle inequality, $\|x'_i - x'_j\| \le \|x'_i - m_{ij}\| + \|m_{ij} - m\| + \|m - m_{ji}\| + \|m_{ji} - x'_j\| \le \frac{1}{2}(r_{ij} + r_{ji})$. The segment $[x'_i, x'_j]$ lies within $\|x'_j - \hat{x}_j\| \le \delta_{ij}$ of $[x'_i, \hat{x}_j]$, so every obstacle crossing it, un-inflated, crosses $[x'_i, \hat{x}_j]$ when inflated by δ_{ij}: $n(x'_i, x'_j) \le n^{\delta_{ij}}(x'_i, \hat{x}_j)$ and, R_ε being non-increasing in n, $R_\varepsilon(n(x'_i, x'_j)) \ge r_{ij}$. Likewise $R_\varepsilon(n(x'_i, x'_j)) \ge r_{ji}$. Hence $\|x'_i - x'_j\| \le \max(r_{ij}, r_{ji}) \le R_\varepsilon(n(x'_i, x'_j))$. □

Corollary 1 (Connectivity during delivery)**.** *If all UAVs of a connected component of $\mathcal{G}_\varepsilon(t)$ containing the responder are tethered, take steps satisfying* (7)*, and no UAV fails, that component remains connected at $t + \Delta t$.*

Remark. Proposition 1 is a one-step guarantee for feasible steps; it does not by itself make (7) feasible at every step (the inflated obstruction count can shrink r_{ij} even for a hovering UAV). We therefore also report the empirical rate of preserved-link violations in all experiments, and a Monte-Carlo check of the obstacle-free synchronous case (10^4 random swarms of 2–50 UAVs, 25 adversarial steps each, no violation).

Proof. By Lemma 2 the mutually selected edges span the component, and by Proposition 1 each of them is reliable at $t + \Delta t$ (the responder never moves, so a one-sided constraint suffices for its edges). □

Proposition 2 (Probabilistic delivery path)**.** *Under Corollary 1, the probability that, in a slot, the realised (fading) network restricted to the preserved spanning tree $\mathcal{M}$ is disconnected satisfies* $\Pr[\text{disc.}] \le \sum_{e \in \mathcal{M}} P_{\text{out}}(e) \le (|\mathcal{M}|)\,\varepsilon$.

Proof. The realised tree is disconnected only if at least one of its edges is in outage; apply the union bound and $P_{\text{out}}(e) \le \varepsilon$ for preserved edges. □

D. Relay tether

Proposition 1 needs *both* endpoints tethered, but the relays between a delivering UAV and the responder are usually exploring. A delivering, connected UAV therefore stamps $t_{\text{tether}} = t + T_{\text{rel}}$ in its beacon; stamps propagate by maximum, and a connected UAV is tethered while $t \le t_{\text{tether}}$. The whole connected component thus holds its links during a delivery and is released T_{rel} after it ends; because the stamp is set only by delivering UAVs, tethers cannot sustain each other indefinitely.

E. Information exposure

Proposition 3 (Exposure bound)**.** *Assume the creator of a confirmed entry stays alive until it is secured or fails, flies at speed v along paths at most $\kappa \ge 1$ times the straight-line distance, and that delivery completes within τ_c of reaching the responder's connected component. Then the entry is unsecured for at most $W = T_h + \kappa D/v + \tau_c$, with $D = \max_{x \in \mathcal{A}} \|x - x_0\|$. If the UAVs holding it fail independently at rate λ_f and the entry has k holders during that window,*

$$\Pr[\textit{entry lost}] \le \left(1 - e^{-\lambda_f W}\right)^k \le (\lambda_f W)^k. \tag{8}$$

Proof. By (4), DELIVER starts at most T_h after confirmation; the return takes at most $\kappa D/v$ and the hand-off at most τ_c, after which $0 \in H_e$ and the entry is secured (the responder does not fail). It is lost only if every holder fails within the window, which has probability $(1 - e^{-\lambda_f W})^k$ under independence; $1 - e^{-z} \le z$. □

Proposition 3 makes the role of T_h explicit: it bounds the exposure of confirmed information to UAV failures (the information-loss side), while larger T_h leaves more time for search (the discovery side). Section VI measures the resulting trade-off and compares the measured losses with (8).

F. Collaborative synthetic-aperture sensing

A tentative entry ($c_{\text{lo}} \le c_e < c_{\text{hi}}$) requests a second view from the nearest neighbour at a lateral offset of $\pm b$ ($b = 15\,\text{m}$) perpendicular to the observer–target direction, admitted only if the flank point keeps that neighbour's preserved links feasible. The flanking view enters the corroborative fusion below.

G. Corroborative multi-view fusion

A victim is observed by several UAVs from different directions. For each entry the ledger keeps the best confidence $c_{e,v}$ per viewing sector v (eight azimuth sectors around the entry), so repeated frames from one viewpoint are not counted twice. The textbook naive-Bayes rule $\operatorname{logit} c_e = \sum_v \operatorname{logit} c_{e,v}$ treats every low-confidence view as evidence *against* a victim. Under occlusion this is wrong: a victim partly covered by a slab returns weak detections from the occluded side and strong ones from the open side, and a weak view from a blocked angle says little about absence. With many UAVs the many weak views outvote a clear view and the victim is never confirmed. We therefore fuse corroboratively,

$$\operatorname{logit} c_e = \max_v \operatorname{logit} c_{e,v} + \sum_{v \ne v^\star} \left[\operatorname{logit} c_{e,v}\right]_+, \tag{9}$$

where $v^\star$ is the best view: the best view sets the confidence, other views can only add corroborating evidence, and a single view keeps its own confidence. Equation (9) is the component with the largest measured effect in dense swarms (Section VI-H); its cost is that two independent false alarms on the same spot corroborate each other, which we evaluate with deliberately adverse clutter (Section VI-G).

H. Complexity

Per planning step a UAV evaluates P particles for I iterations; each evaluation costs $O(k)$ for $k = |\mathcal{S}_i|$ outage look-ups plus an $O(1)$ map look-up (the footprint convolution is computed once per step, $O(M)$ for M map cells). The local MST costs $O(k_h^2 \log k_h)$ for the k_h neighbours heard. None of these terms depends on N; the measured per-agent compute time versus N is reported in Section VI.

Algorithm 1 SENTINEL agent i (every slot Δt)

sense: record look, add geolocated detections to the ledger (fused per viewpoint)
broadcast beacon $(x_i, \text{ledger}, \text{acks}, \text{heartbeat}, t_{\text{tether}})$; merge received beacons
if planning slot **then**
    update mode with (4); issue SA flank requests for own tentative entries
    if DELIVER and not connected **then**
        waypoint $\leftarrow x_0$
    else
        waypoint $\leftarrow \arg\max_{x \in \mathcal{D}_i} F_i(x)$ by PSO ▷ $S_i \equiv 0$ unless tethered
    end if
end if
$u_i \leftarrow$ step toward waypoint (rotated away from $\mathcal{O}$)
if tethered **then**
    $u_i \leftarrow \alpha^\star u_i$ with $\alpha^\star$ from (7)
end if
fly $x_i \leftarrow x_i + u_i$

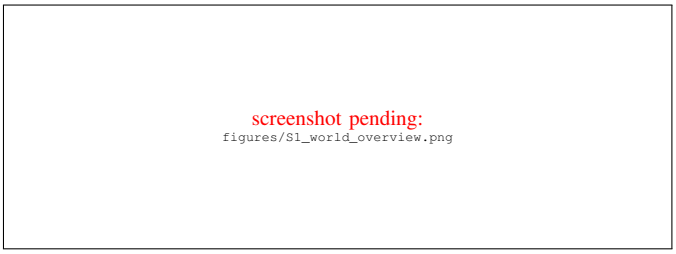


Fig. 1. Generated 100 m disaster area in Gazebo Harmonic (top view): building shells, rubble, warm debris and victims.

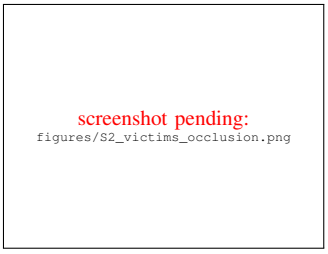




Fig. 2. Left: victims partly and fully covered by concrete slabs. Right: nadir thermal frame (white-hot) from 30 m.

V. IMPLEMENTATION

a) One algorithm library, two execution environments: All algorithms of Section IV (channel model, ledger, fusion, PSO, link preservation, delivery trigger) are implemented once in a ROS-independent Python library. The same library is executed by (i) a fast multi-agent simulator used for the statistical evaluation and (ii) ROS 2 Humble nodes that fly PX4 multicopters in Gazebo Harmonic, so the results of both environments refer to identical decision code.

b) Flight stack: Each UAV is a PX4 x500 quadrotor with a nadir thermal camera (640×512, 50° field of view, 10 Hz, radiometric 16-bit output converted to white-hot 8-bit on board). PX4 runs as software-in-the-loop; the Micro XRCE-DDS agent bridges its uORB topics to ROS 2. Per UAV, four nodes run on the companion computer (Fig. 3): `flight_commander` (offboard position control and world-frame pose), `thermal_vision` (YOLO26n detection and geolocation of every box into the world frame from the camera model, pose and heading), `emps_agent` (the SENTINEL agent) and the camera bridge. There is no central coordinator.

c) Radio emulation and failures: All inter-UAV beacons pass through a `comm_emulator` node that delivers each packet only if its sampled SINR, computed with the channel model of Section III and the building geometry of the world, exceeds the decoding threshold; it also hosts the responder radio. A `chaos_monkey` node injects crashes (force-disarm), radio cuts and brownouts on a schedule, and a `metrics_logger` computes the metrics of Section VI from ground truth. A beacon carries the sender's position, ledger, acknowledgements and heartbeat; it averages 4.1 kB (20 kB when the search map is included every fourth beacon).

d) Disaster scenes: Worlds are generated procedurally from a seed: tall building shells (no-fly, link-obstructing), low rubble that occludes victims, warm debris (316 K) that causes false alarms, and Rescue Randy mannequins with a 310 K body heat signature, partly or fully covered by concrete slabs (Figs. 1–2). The same seed produces the same scene in Gazebo and in the fast simulator, which is what makes the cross-check of Section VI-J possible.

e) Perception in the loop: Fig. 4 shows YOLO26n running on the Gazebo thermal feed during flight. Because the detector is trained on real thermal imagery, the rendered scenes produce false boxes on rubble edges and missed mannequins; we therefore evaluate detection accuracy on HIT-UAV (Section VI-B), and the cross-check of Section VI-J uses the calibrated detector model of the fast simulator, so that both environments see the same perception.

f) Reproducibility: A single command starts the complete Gazebo/PX4 demonstration; every experiment is defined by a configuration file, runs deterministically from its seed, and the tables and figures in this paper are generated by script from the stored results. The source code, scenes, configurations and raw results are released with the paper.

VI. EVALUATION

A. Methodology

a) Simulators: All algorithms are implemented once, in a ROS-independent library that is executed both by a fast multi-agent simulator and by the ROS 2 nodes running against PX4 SITL in Gazebo (Section V). The fast simulator advances all UAVs with a 0.5 s control step, samples every broadcast through the channel model of Section III (path loss, building obstructions, correlated shadowing, Rician/Rayleigh fading, collision interference) and produces detections with the calibrated sensing model. Each run starts from a procedurally generated disaster area (tall buildings that are no-fly zones and obstruct links, low rubble that occludes victims, warm debris that causes false alarms, 25 victims of which one is fully buried) and lasts 600 s. Unless stated otherwise the area is 400 m×400 m, $N = 8$, the channel is urban-rubble, and UAVs fail as a Poisson process with rate 0.01 s^{-1} per swarm.

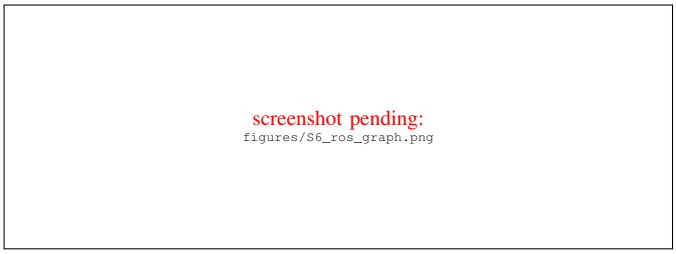
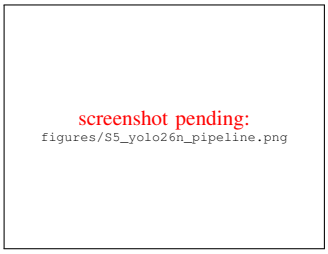


Fig. 3. ROS 2 node graph with six UAVs: per-UAV perception, agent and flight nodes communicate only through the radio emulator; no central coordinator.



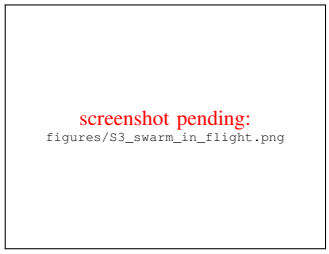


Fig. 4. Left: YOLO26n boxes on the Gazebo thermal feed (green: confirmed band, orange: SA band); the rendering gap to real thermal imagery is visible. Right: six UAVs searching.

b) Metrics: The primary metric is the share of all victims whose report reaches the responder within 60 s of their first confirmation (*reported* $\leq$*60 s*); its denominator is the number of victims in the area, so a method cannot score well by detecting few. We also report the share of victims reported by the end of the mission, the 90th-percentile report latency, coverage of the reachable area, information loss (victims confirmed by some UAV whose every holder failed before the report reached the responder), false confirmed entries, and the per-agent planning time.

c) Baselines: B0: constriction PSO with the same exploration layer and no communication term; B1: the earlier SENTINEL design (nearest-neighbour distance spring, $R = 50$ m); B2: communication-aware PSO with a soft outage penalty on all neighbours; B3: algebraic-connectivity gradient control [2] given the *exact* λ_2 and Fiedler vector of the swarm at every step (an oracle that favours B3); B4: edge-tension potential with hysteresis [4]; B5: MAPPO [17] with a parameter-shared actor on local observations and a centralised critic; B6: boustrophedon sweep of static sectors. All methods share sensing, gossip, the ledger and the channel, so they differ only in how UAVs move.

d) Fair tuning and statistics: Every method's free parameters were tuned with the same budget (Optuna TPE, 30 trials, each scored on six runs in an urban $N = 8$ and a severe $N = 4$ scenario) on seeds 1000–1002, which are never used for evaluation. MAPPO was trained for 600 PPO iterations of 16 episodes each (9 600 missions) on separate seeds and selected on the same tuning seeds; its training return plateaus after about 150 iterations (Fig. 5), so the reported policy is not limited by training length. Each configuration is evaluated on seeds 0–29; all methods see the same area, victims and failure schedule for a given seed (common random numbers). We report means with 95 % bootstrap confidence

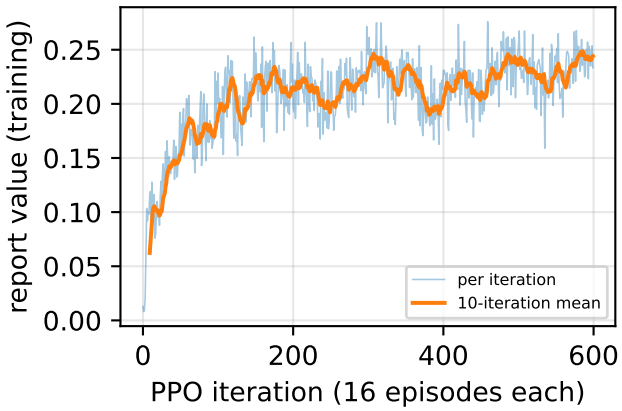


Fig. 5. Training curve of the MAPPO baseline (B5): time-discounted report value per PPO iteration and its 10-iteration mean. The return plateaus after about 150 iterations.

intervals and test SENTINEL against every other method with paired Wilcoxon signed-rank tests, Holm-corrected within each scenario (marked * in the tables for $p < 0.05$). In total the evaluation comprises about 11 000 simulated missions.

B. Onboard perception

Table III compares YOLO26n with YOLOv8n trained under an identical protocol on HIT-UAV [1]. The two detectors reach the same mAP50 (0.948 vs. 0.947); YOLO26n is more precise (0.932 vs. 0.909) and better localised (mAP50-95 0.533 vs. 0.520) with slightly lower recall, and it needs 19 % fewer parameters and 31 % fewer operations. On a desktop GPU the two are equally fast (about 1.8 ms end-to-end with TensorRT FP16); we therefore attribute YOLO26n's suitability for low-power hardware to its lower compute, not to measured speed, and we do not report embedded latency, which was not measured. The accuracy values are the validation metrics recorded by the training runs.

C. Communication conditions

Table IV varies the channel from open terrain to the severe profile. SENTINEL reports the largest share of victims within 60 s in every profile and is significantly better than all seven baselines in the open, suburban and urban settings (e.g. 0.575 vs. 0.475 for the best baseline, B3, in urban rubble). In the severe channel it remains ahead (0.433 vs. 0.377) but the difference to the best baseline is not significant. Its total share of reported victims is 0.56–0.58 across profiles against 0.46–0.50 for the best baseline of each profile. In the severe profile SENTINEL's 90th-percentile latency (99 s) is longer than B0's (69 s): it delivers more of the reports that would otherwise never arrive, and some of these arrive late.

D. Failures

Tables V and VI apply six failure models: none, a single crash, 30 % of the swarm crashing simultaneously, three cascading crashes, Poisson crashes at twice the default rate, and a

TABLE III
THERMAL DETECTORS TRAINED WITH THE SAME PROTOCOL (HIT-UAV, 100 EPOCHS, ADAMW, 640 PX, SEED 0). ACCURACY: VALIDATION METRICS RECORDED BY THE TRAINING RUN. LATENCY: END-TO-END P50 ON AN RTX 3060 (DESKTOP), PYTORCH FP32 AND TENSORRT FP16.

Model	Params [M]	GFLOPs	mAP50	mAP50-95	P	R	PyTorch [ms]	TensorRT [ms]
YOLO26n	2.53	6.0	0.948	0.533	0.932	0.900	3.72	1.81
YOLOv8n	3.14	8.7	0.947	0.520	0.909	0.913	3.32	1.73

TABLE IV
VICTIMS REPORTED WITHIN 60 S PER CHANNEL PROFILE (N=8, 400 M, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	open	severe	suburban	urban
SENTINEL	**0.577**	**0.433**	**0.560**	**0.575**
B0 PSO	0.473*	0.377	0.456*	0.453*
B1 Earlier design	0.457*	0.373	0.427*	0.409*
B2 Comm-PSO	0.472*	0.336*	0.371*	0.468*
B3 λ_2-grad.	0.477*	0.353	0.453*	0.475*
B4 Potential	0.501*	0.300*	0.099*	0.127*
B5 MAPPO	0.405*	0.333*	0.421*	0.396*
B6 Lawnmower	0.424*	0.373	0.423*	0.421*

gradual transmit-power brownout. In the urban channel SENTINEL is significantly better than every baseline under every failure model, by 23–30 % over the best baseline. In the severe channel it is ahead of or tied with the best baseline (often the lawnmower sweep), with no significant loss. Information loss, i.e. confirmed victims whose every holder failed before reporting, is small for *all* methods: at most 0.9 % averaged over all failure models and at most 8.5 % under high-rate Poisson crashes in the severe channel (3.2 % for SENTINEL). Gossip replication between UAVs that meet during the search protects most information, so we make no claim that SENTINEL reduces loss compared with the baselines.

E. Scalability

Tables VII and VIII vary the swarm from 3 to 50 UAVs. On the 400 m area SENTINEL is significantly better than all baselines at every swarm size, from 0.203 vs. 0.153 with three UAVs to 0.788 vs. 0.725 with fifty. On the smaller 200 m area, where contacts are frequent, it has the highest mean at every swarm size but is significantly better than all baselines only for $N = 50$ (0.809 vs. 0.779). In dense swarms MAPPO (B5) becomes the strongest baseline (0.691–0.779 for $N \geq 10$ on the 200 m area), while it remains well behind in sparse settings. The per-agent planning time grows with the number of neighbours a UAV hears, not with N directly: from 4.0 to 11.1 ms on the dense 200 m area and from 5.1 to 9.5 ms on the 400 m area as N grows from 3 to 50 (Fig. 6), far below the 1 s planning period. The all-neighbour communication-aware PSO (B2) grows faster, to 20.6 ms.

F. Moving obstacles

With 5 or 15 tall debris objects drifting through the area, blocking links and hiding victims (Table IX), SENTINEL keeps reporting 0.58 of the victims within 60 s and remains significantly better than all baselines (best baseline 0.48–0.51).

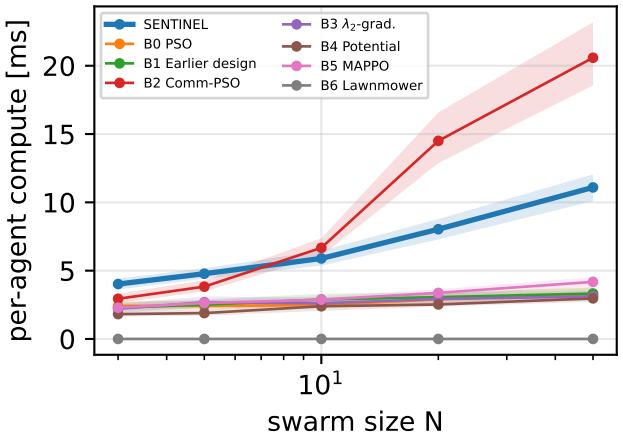


Fig. 6. Per-agent planning time vs. swarm size on the 200 m area (mean and 95 % CI over 30 seeds).

G. Detection uncertainty

We perturbed the calibrated detector model around its default: confidence noise $\sigma \in \{0.04, 0.08, 0.15, 0.25\}$, a weaker detector ($c_{\max} \in \{0.8, 0.7\}$) and more frequent false alarms. SENTINEL stays significantly ahead of B0 and B6 in all eight settings (0.45–0.81 vs. at most 0.59). Because the default false-alarm confidences rarely exceed the confirmation threshold, we also ran a deliberately adverse case (Table X): hot clutter that scores up to a mean of 0.55, the confidence noise applied to false alarms as well, and false alarms three times more frequent. SENTINEL still reports 0.71–0.73 of the victims within 60 s against at most 0.50 (B0), with 0.18 false confirmed entries per mission on average (B0: 0.29). Moving the SA band and the confirmation threshold (0.3/0.6 to 0.4/0.85) changes the on-time share between 0.64 and 0.74 without producing false confirmations.

H. What produces the gains

Table XI removes one component at a time. Two components matter. First, multi-view fusion: removing it reduces on-time reports significantly (urban, $N = 8$: 0.519 vs. 0.575), and in the dense swarm of Table XII the corroborative rule is decisive (0.809, against 0.656 with plain naive-Bayes fusion and 0.720 for B0). Second, the delivery trigger: Fig. 7 sweeps T_h; in the severe channel with eight UAVs, delivering immediately ($T_h = 0$) raises the on-time share from 0.388 (never

TABLE V
VICTIMS REPORTED WITHIN 60 S UNDER FAILURE MODELS (URBAN-RUBBLE CHANNEL, N=8, 400 M). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	brownout	cascading	none	Poisson	30	
SENTINEL	**0.687**	**0.643**	**0.691**	**0.459**	**0.675**	**0.689**
B0 PSO	0.539*	0.509*	0.537*	0.344*	0.513*	0.523*
B1 Earlier design	0.513*	0.495*	0.501*	0.356*	0.499*	0.501*
B2 Comm-PSO	0.555*	0.524*	0.560*	0.340*	0.521*	0.545*
B3 λ_2-grad.	0.532*	0.489*	0.533*	0.339*	0.505*	0.523*
B4 Potential	0.183*	0.161*	0.193*	0.085*	0.144*	0.160*
B5 MAPPO	0.489*	0.457*	0.499*	0.328*	0.453*	0.473*
B6 Lawnmower	0.521*	0.488*	0.521*	0.345*	0.485*	0.511*

TABLE VI
VICTIMS REPORTED WITHIN 60 S UNDER FAILURE MODELS (SEVERE CHANNEL, N=8, 400 M). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	brownout	cascading	none	Poisson	30	
SENTINEL	**0.517**	0.452	**0.529**	**0.340**	**0.496**	**0.512**
B0 PSO	0.444	0.403	0.456	0.291	0.417	0.440
B1 Earlier design	0.448	0.423	0.469	0.295	0.440	0.452
B2 Comm-PSO	0.427*	0.385	0.439*	0.259*	0.400*	0.428
B3 λ_2-grad.	0.380*	0.363*	0.409*	0.249*	0.392*	0.409*
B4 Potential	0.359*	0.309*	0.396*	0.200*	0.359*	0.361*
B5 MAPPO	0.431*	0.383	0.445*	0.267*	0.412*	0.440*
B6 Lawnmower	0.499	**0.453**	0.508	0.287	0.468	0.495

TABLE VII
VICTIMS REPORTED WITHIN 60 S VS SWARM SIZE N (400 M MAP, URBAN, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	N=3	N=5	N=10	N=20	N=50
SENTINEL	**0.203**	**0.389**	**0.667**	**0.751**	**0.788**
B0 PSO	0.153*	0.285*	0.535*	0.619*	0.656*
B1 Earlier design	0.109*	0.232*	0.499*	0.647*	0.707*
B2 Comm-PSO	0.147*	0.313*	0.535*	0.641*	0.685*
B3 λ_2-grad.	0.136*	0.273*	0.529*	0.636*	0.665*
B4 Potential	0.017*	0.027*	0.240*	0.485*	0.712*
B5 MAPPO	0.153*	0.273*	0.503*	0.628*	0.725*
B6 Lawnmower	0.151*	0.265*	0.444*	0.493*	0.619*

TABLE VIII
VICTIMS REPORTED WITHIN 60 S VS SWARM SIZE N (200 M MAP, URBAN, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	N=3	N=5	N=10	N=20	N=50
SENTINEL	**0.475**	**0.620**	**0.720**	**0.743**	**0.809**
B0 PSO	0.457	0.597	0.660*	0.697*	0.720*
B1 Earlier design	0.360*	0.565	0.652*	0.692*	0.724*
B2 Comm-PSO	0.459	0.579	0.647*	0.725	0.732*
B3 λ_2-grad.	0.345*	0.560	0.664*	0.695*	0.727*
B4 Potential	0.044*	0.112*	0.471*	0.656*	0.735*
B5 MAPPO	0.379	0.559	0.691	0.729	0.779*
B6 Lawnmower	0.419	0.536*	0.541*	0.641*	0.703*

delivering) to 0.497 and lowers the 90th-percentile latency from 97 s to 34 s, while the total share of reported victims stays at 0.50. In the urban channel the same trend is present but not significant. The remaining components (outage-based instead of distance-based spring, LMST instead of nearest-neighbour preserved set, information-dependent tension, projection) do

TABLE IX
VICTIMS REPORTED WITHIN 60 S WITH MOVING OBSTACLES (URBAN, N=8). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	0 moving	5 moving	15 moving
SENTINEL	**0.575**	**0.585**	**0.579**
B0 PSO	0.453*	0.481*	0.460*
B1 Earlier design	0.409*	0.424*	0.447*
B2 Comm-PSO	0.468*	0.489*	0.509*
B3 λ_2-grad.	0.475*	0.479*	0.487*
B4 Potential	0.127*	0.132*	0.141*
B5 MAPPO	0.396*	0.449*	0.431*
B6 Lawnmower	0.421*	0.443*	0.425*

TABLE X
VICTIMS REPORTED WITHIN 60 S WITH HOT-CLUTTER FALSE ALARMS (URBAN, N=8, σ=0.15). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	μ_{fp}=0.22, r=0.1	μ_{fp}=0.4, r=0.1	μ_{fp}=0.55, r=0.1	μ_{fp}=0.55, r=0.3
SENTINEL	**0.731**	**0.719**	**0.724**	**0.709**
B0 PSO	0.501*	0.491*	0.500*	0.493*
B6 Lawnmower	0.480*	0.480*	0.479*	0.481*
no SA	0.715	0.693	0.708	0.668
no fusion	0.672*	0.655*	0.676*	0.691

not change the outcome significantly, and the SA flank is not significant in the tuned configuration. These mechanisms act only during the short delivery tethers; we regard them as the safety layer examined next, not as the source of the performance gains. Keeping the swarm connected at all times (SENTINEL-strict) lowers on-time reporting to 0.03–0.13.

I. Link preservation during delivery

Proposition 1 is a one-step guarantee for feasible steps. Over all experiments we counted the tethered, mutually preserved

TABLE XI
ABLATION: VICTIMS REPORTED WITHIN 60 S (400 M, POISSON FAILURES). MEAN OVER 30 SEEDS; BEST PER COLUMN IN BOLD; *: DIFFERS FROM SENTINEL (PAIRED WILCOXON, HOLM-CORRECTED, $p < 0.05$).

Method	severe, N=4	severe, N=8	urban, N=4	urban, N=8
SENTINEL	**0.212**	0.433	0.313	0.575
SENTINEL-strict	0.029*	0.061*	0.064*	0.131*
distance spring	0.200	0.433	0.307	0.561
NN set	**0.212**	0.433	0.313	0.573
no info. tension	0.211	0.433	0.312	0.572
no projection	0.200	0.435	0.308	0.545
no SA	0.189	**0.455**	**0.316**	**0.583**
no fusion	0.187	0.408	0.288	0.519*
naive-Bayes fusion	0.167	0.439	0.300	0.540

TABLE XII
DENSE SWARM (200 M, N=50): VICTIMS REPORTED WITHIN 60 S.

Method	50
SENTINEL	**0.809**
B0 PSO	0.720*
no SA	0.799
no fusion	0.740*
naive-Bayes fusion	0.656*
no SA, no fusion	0.719*

links that were reliable before a step and unreliable after it: 15 violations in 436 057 link-steps (3.4×10^{-5}); with the swarm tethered at all times (SENTINEL-strict) the rate is 2.6×10^{-3}. The residual violations come from steps where the constraint was infeasible and the UAV hovered (Section IV-C). Information loss did not vary consistently with T_h at the failure rates studied (at most 6.2 % in any cell), so the exposure bound of Proposition 3 could not be confirmed or refuted empirically; it remains a design guarantee.

J. Cross-validation in PX4/Gazebo and the earlier scenario

To check that the fast simulator does not depend on idealised flight, we flew SENTINEL and B0 with three PX4 x500 multicopters in Gazebo Harmonic on the same three 100 m areas and seeds (300 s, crash of one UAV at 150 s), with the same agent code, tuned values, channel emulation and detector model. The share of reported victims (SENTINEL 0.53 vs. 0.60 in the fast simulator; B0 0.60 vs. 0.62), the packet delivery ratio (0.79–0.81 vs. 0.80–0.83) and the connected fraction agree within 0.07; coverage of SENTINEL is higher in SITL (0.99 vs. 0.83). With three seeds this is a consistency check rather than a statistical comparison. Finally, re-running the scenario of the earlier SENTINEL design (25 m area, three UAVs, one crash) shows that it cannot separate the methods: the area is far smaller than the radio range, every method stays connected throughout, and 64–69 % of the victims are reported by all of them. The remaining victims are either buried or so close to another victim (4–5 m) that their reports merge (7.8–11 % of confirmations), which is a limitation of the association gate rather than of connectivity.

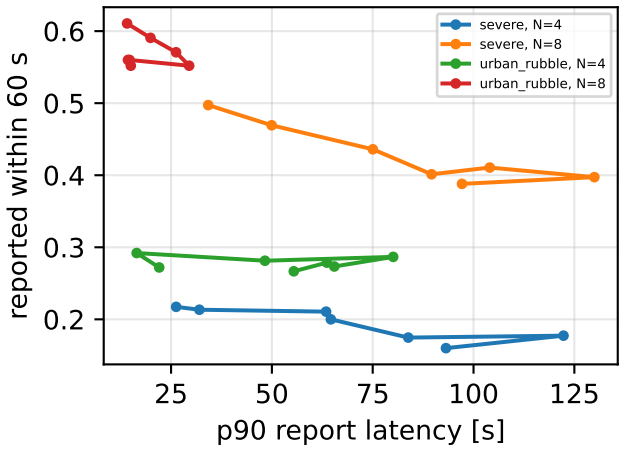


Fig. 7. Delivery trigger $T_h \in \{0, 15, 30, 60, 120, 240, \infty\}$: on-time share vs. 90th-percentile report latency for four regimes (each point: mean of 30 seeds).

VII. DISCUSSION AND LIMITATIONS

a) Where the gains come from: The experiments support a narrower claim than "keeping the network alive". Connectivity in itself is not the bottleneck in the scenarios we studied: keeping the swarm connected at all times (SENTINEL-strict, B3, B4) limits the search and lowers on-time reporting, and gossip between UAVs that meet during the search already protects most information from failures. What raised timely reporting was deciding *when* a UAV should stop searching to deliver what it found, and combining views so that occlusion does not hide confirmed victims. The link-preservation layer (Proposition 1) did not change performance measurably; its role is to make deliveries reliable when they happen.

b) Choice of T_h: T_h trades on-time reporting against total discovery. The value tuned on our mixed objective (116 s) is not the best for a 60 s deadline in the severe channel, where $T_h = 0$ was significantly better. In practice T_h should follow the responders' reporting deadline; adapting it online to the observed contact rate is a natural extension.

c) Tuning sensitivity: An earlier configuration of SENTINEL that also tethered UAVs softly while exploring helped dense swarms but reduced coverage in the severe channel; the evaluated design does not use this tether. Results of that configuration are released with the others.

d) Limitations: (i) All results are from simulation: a fast multi-agent simulator and PX4 software-in-the-loop in Gazebo; no hardware flights were performed. (ii) The channel model is parameterised from published ranges and has not been calibrated against measurements in rubble. (iii) Detection accuracy is reported from the validation split recorded by the training runs; the detector transfers poorly to rendered thermal scenes, which is why the in-the-loop comparison uses a calibrated detector model. (iv) Victims closer than the association gate (about 5 m) can be merged into one report. (v) MAPPO was trained to a plateau (600 iterations) with one observation and reward design; other MARL formulations may perform better, in particular in dense swarms, where it is already the

strongest baseline. (vi) The exposure bound (Proposition 3) could not be tested empirically because losses were rare at the failure rates studied. (vii) Latency was measured on a desktop GPU, not on embedded hardware.

VIII. Conclusion

We studied how a UAV swarm without infrastructure can get victim reports to responders in time. SENTINEL lets UAVs search freely, triggers delivery from the information each UAV has confirmed, fuses views so that occluded angles cannot veto a clear detection, and preserves the links of a delivery path for every feasible step (Proposition 1). In about 11 000 simulated missions against seven tuned baselines, it reported significantly more victims within 60 s than every baseline in most settings (urban, suburban and open channels, all failure models in urban rubble, swarms of 3–50 UAVs on the 400 m area, moving obstacles and detector uncertainty); in the severe channel it was ahead of or tied with the best baseline, and it was never significantly worse. A PX4/Gazebo cross-check agreed with the fast simulator within 0.07. The ablation located the gains in the delivery trigger and the fusion rule rather than in continuous connectivity. Future work will adapt the delivery trigger to the observed contact rate, calibrate the channel model with measurements, and validate the system in hardware flights.

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